

Phishing Email Analysis Report

Identifying and Analyzing IOCs

Prepared by	Muhammad Umer
Date	August 10, 2026
Lab Environment	Ryzen 7 Pro 5850U, 16GB RAM, 512GB NVMe SSD
Hypervisor	Oracle VirtualBox
VM OS	Windows 10
Report Version	v1.0

Table of Contents

1. Executive Summary.....	3
Key Achievements.....	3
2. Lab Architecture & Environment	4
2.1 Hardware	4
2.2 Software Stack	4
2.3 Email Sample.....	4
2.4 Email Header Analysis.....	4
2.5 Email Body Analysis.....	6
3. IoCs Summery	7
4. Conclusion.....	7

1. Executive Summary

The key objective of this project is to replicate a real-world phishing email analysis to identify and critically evaluate indicators of compromise. The complete analysis process was performed on a VM to avoid any accidental exposure. The phishing email sample was obtained from the 'phishing_pot' GitHub repository, confirming it as a phishing email. The email header was analyzed to validate sender information and routing path. The extracted IOCs from the email header were enriched using OSINT tools like VirusTotal, AbuseIPDB, and MXToolbox. The OSINT tools did not provide any evidence of malicious activities related to the IPs — the main reason being the age of the email sample. SPF failure, missing DKIM signature, and absence of DMARC policy were the most prominent indicators of a phishing email. The email body highlighted social engineering tactics including urgency, authority, and reward. By summarizing all IOCs, the sample was declared as an Advance Fee Fraud.

Key Achievements

- Learned and applied best practices for safe handling of phishing emails
- Identified and analyzed IOCs (IPs, URLs, SPF authentication result, DKIM signature status, DMARC policy, etc.) from email headers and body
- Gained hands-on experience with OSINT tools for IOC enrichment
- Traced complete email routing path across 7 SMTP hops, identifying compromised relay servers and suspicious time delays
- Developed ability to identify social engineering tactics in phishing emails

2. Lab Architecture & Environment

2.1 Hardware

Component	Specification
Laptop Model	Lenovo ThinkPad T14
Processor	AMD Ryzen 7 Pro 5850U (8 Cores / 16 Threads)
RAM	16GB
Storage	512GB NVMe SSD
OS	Windows 11

2.2 Software Stack

Tool	Purpose
VirtualBox	Hypervisor for VM hosting
Windows Guest (Win 10)	Guest OS for email analysis
VirusTotal	Analyze URLs, IPs, file hashes
AbuseIPDB	Check if IP has been reported for abuse
MXToolBox	Analyze email headers, check blacklists

2.3 Email Sample

Phishing email sample was download from phishing_pot GitHub repository,

https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1006.eml

and .eml file was opened in Notepad++ for analysis.

2.4 Email Header Analysis

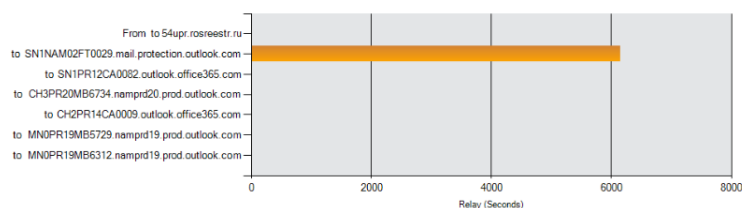
The email header was pasted into the Email Header Analyzer tool of MXToolbox. The email originated from 85.208.139.60 (Germany) and was relayed through 54upr.rosreestr.ru / 109.202.24.52 (Russia), from where it was forwarded to Microsoft Outlook's SMTP server after a 2 hour delay from the originating server — possibly due to a long queue of waiting emails, indicating a mass phishing campaign. After that, the email hopped through 4 internal Microsoft SMTP servers before being delivered to the recipient.

The routing of the email through Germany and Russia appears to be a deliberate technique to conceal the true identity of the attacker, making attribution significantly harder.

54upr.rosreestr.ru (109.202.24.52) is a Russian government official server. The delivery of a phishing email through this server indicates that it was compromised and misused by the threat actor, likely due to outdated and unpatched server software.

Relay Information

Received Delay:	6167 seconds
-----------------	--------------



Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	85.208.139.60	54upr.rosreestr.ru	Microsoft SMTPSVC(6.0.3790.4675)	7/29/2023 8:00:51 PM	✖
2	2 hour	54upr.rosreestr.ru 109.202.24.52	SN1NAM02FT0029.mail.protection.outlook.com 10.97.4.175	Microsoft SMTP Server	7/29/2023 9:43:25 PM	✖
3	9 seconds	SN1NAM02FT0029.eop-nam02.prod.protection.outlook.com 2603:10b6:802:21:cafe::84	SN1PR12CA0082.outlook.office365.com 2603:10b6:802:21::17	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	7/29/2023 9:43:34 PM	✔
4	0 seconds	SN1PR12CA0082.namprd12.prod.outlook.com 2603:10b6:802:21::17	CH3PR20MB6734.namprd20.prod.outlook.com 2603:10b6:610:168::8	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	7/29/2023 9:43:34 PM	✔
5	3 seconds	CH3PR20MB6734.namprd20.prod.outlook.com 2603:10b6:610:60:cafe::6f	CH2PR14CA0009.outlook.office365.com 2603:10b6:610:60::19	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	7/29/2023 9:43:37 PM	✔
6	0 seconds	CH2PR14CA0009.namprd14.prod.outlook.com 2603:10b6:610:60::19	MN0PR19MB5729.namprd19.prod.outlook.com 2603:10b6:208:374::21	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	7/29/2023 9:43:37 PM	✔
7	1 second	MN0PR19MB5729.namprd19.prod.outlook.com ::1	MN0PR19MB6312.namprd19.prod.outlook.com	HTTPS	7/29/2023 9:43:38 PM	✖

Evaluation the following section of email header,

Authentication-Results: spf=fail (sender IP is 109.202.24.52)

smtp.mailfrom=postfiji.com.fj; dkim=none (message not signed)

header.d=none; dmarc=none action=none

header.from=postfiji.com.fj; compauth=fail reason=001

Received-SPF: Fail (protection.outlook.com: domain of postfiji.com.fj does not designate 109.202.24.52 as permitted sender) receiver=protection.outlook.com; client-ip=109.202.24.52; helo=54upr.rosreestr.ru;

In the second hop, email was delivered from 54upr.rosreestr.ru (109.202.24.52) to Microsoft Outlook SMTP server on behalf of postfiji.com.fj (rosreestr.ru spoofed as postfiji.com.fj). Microsoft SMTP server validated the Sender Policy Framework (SPF) record of postfiji.com.fj to check either IP 109.202.24.52 is authorized to send email on behalf of postfiji.com.fj. SPF record failed, a red flag.

DKIM signal indicates the authenticity of email received. Message was not signed so Microsoft SMTP server couldn't validate the authenticity of email message using DKIM, another red flag.

On the basis of SPF authentication result and DKIM signature status, DMARC decides what action to take with email message (reject/quarantine/none). Microsoft SMTP server checked the DNS record of postfiji.com.fj and couldn't find the DMARC policy record. As a

result, after hopping through 4 internal Microsoft 365 servers, email was delivered to recipient.

Moving ahead,

Another very prominent IOC was the identical sender and receiver email address. The objective of this deception was to confuse the recipient and trick them into responding. This technique is known as "Same Sender Spoofing".

The true attacker contact was revealed in the Reply-To field: **mywoodforestbiz.7@gmail.com**, a free Gmail account, indicating the attacker wanted replies directed to a completely different address than the spoofed from address.

2.5 Email Body Analysis

In the phishing email body, mainly three social engineering tactics were observed:

Authority: People are conditioned to respect and obey authority, that's why the sender appears to be a diplomatic agent to make the email appear credible and actionable.

Reward: People usually respond if they feel they can get a reward. As a result, it is mentioned in the email 'consignment box of \$10.5 million USD', an unavoidable temptation for the victim.

Urgency and Fear: People share their information if they are told they have very little time, or if they fear losing the reward if they don't reply quickly. That's why the statement '**Please hurry up as I do not want to deliver to a wrong destination area**' was added in the email body.

Overall, the objective of the sender is to obtain the victim's phone number and home address, initiate a conversation, and trap the victim into sending advance fees to claim the prize money. This is a classic Advance Fee Fraud, one of the oldest and most profitable social engineering attacks.

3. IoCs Summery

Email Sections	IoC	Description
Email Header	Additional suspicious hops	Identified by analyzing email header using MXToolBox that email was delivered to from Germany (originating IP location) to Russian Government compromised server. The main purpose appears to be obfuscating the true identity of the sender.
	2 hour delay	From the Russian government server, email was forwarded to the victim's mail server after a 2 hour delay, indicating a long queue of emails and possibly a well-organized mass phishing campaign.
	Email spoofing	rosreestr.ru was spoofed as postfiji.com.fj server
	Same Sender Spoofing	Identical sender and receiver email address. The objective of this deception was to confuse the recipient and trick them into responding.
	Failed SPF	54upr.rosreestr.ru (109.202.24.52) was not authorized to send email on the behalf of postfiji.com.fj, email spoofing revealed.
	DKIM None	Email message was not cryptographically signed, authenticity of email content unverifiable
Email Body	Social Engineering Traps	Techniques of authority, reward, and urgency observed in email body to trap the victim into initiating a conversation

4. Conclusion

A real-world phishing email was deeply analyzed in this project to gain familiarity with common indicators of compromise and to gain hands-on experience as a security analyst. The phishing email sample was obtained from the phishing_pot GitHub repository. OSINT tools like VirusTotal and AbuseIPDB diagnosed the originating IP and first hop IP as non-suspicious. The main reason was the aged email sample and the databases of OSINT tools may have been purged or updated since. IOCs like additional suspicious delivery hops, 2 hour delivery delay, email spoofing, failed SPF, missing DKIM, Same Sender Spoofing, and social engineering tactics were observed in the email. By summarizing all these IOCs, this email can clearly be classified as a phishing email, specifically an **Advance Fee Fraud**, where the attacker's objective was to harvest the victim's personal details and ultimately trap them into paying advance fees to claim a non-existent reward.